BIT 4107 – Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

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Admission Number: BIT/2024/59761

School/Faculty: Computing And Informatics

Programme: Bachelor of Information Technology

Academic Year: Year 3

Semester: 2ND

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Week 1: Introduction To Mobile Application Development

Practical Activities Undertaken:

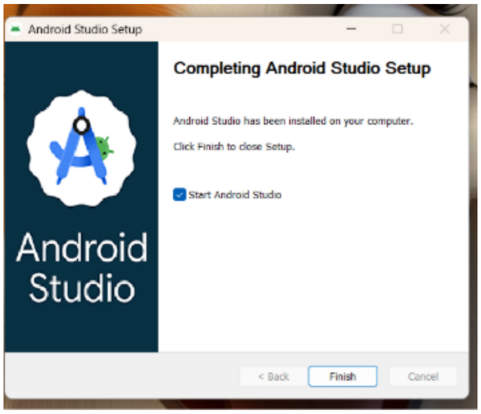

Figure 1.1: Android Studio Setup

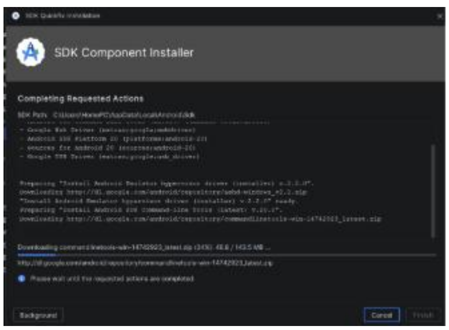

Figure1.2: SDK platforms build

Downloaded the Flutter SDK and initialized the Android Studio installation package. Worked through the initial IDE setup wizards, configured basic SDK command-line dependencies, and mapped out the initial project folders for the upcoming application.

Tools/Software Used:

Android studio installer, flutter SDK, windows file explorer.

Personal Reflection:

During the first week of developing my Flutter Student Management System, I focused on setting up the development environment, understanding the project requirements, and creating the basic structure of the application. I successfully installed and configured Flutter, Android Studio, and other necessary tools, which enabled me to begin designing the user interface. I worked on developing initial screens and explored Flutter widgets, navigation, and layout design. I also started learning how Firebase can be integrated into the project for data storage and management.

Week 2: Mobile Development Environment

Practical Activities Undertaken:

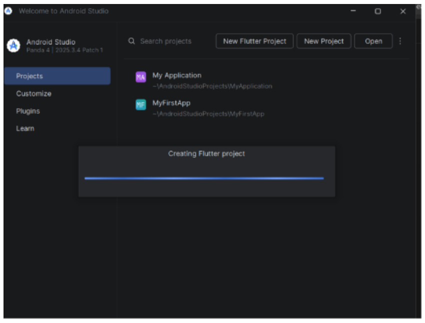

Figure 1.3: Generating flutter framework

Figure 1.4: Generated flutter project inside Android studio

I set up Environmental system variables to connect the Flutter SDK worldwide. created and initialized the AfyaFlow project repository template in Android Studio. reviewed the project architecture files that were automatically generated, made changes to the baseline main.dart script to create a clean environment, and successfully ran a test run on the target virtual device emulator to confirm that the template application compiled error-free.Tools/Software Used:

Android Studio IDE, Flutter SDK, Dart SDK, Command Prompt terminal, Android Virtual Device (AVD) emulator.

Personal Reflection:

During the second week of developing my Flutter Student Management System, I concentrated on implementing the core functionalities of the application. I designed and developed features for adding, viewing, editing, and deleting student records while improving the overall user interface for better usability. I also worked on integrating Firebase services .

Week 3: User Interface Design

Practical Activities Undertaken:

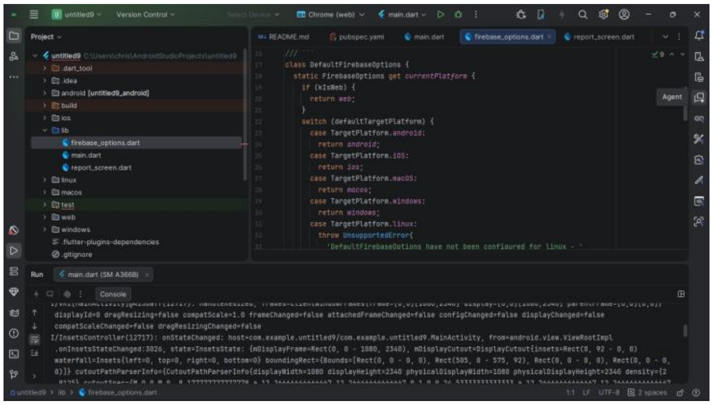

Figure 1.5: Patient screen and GUI

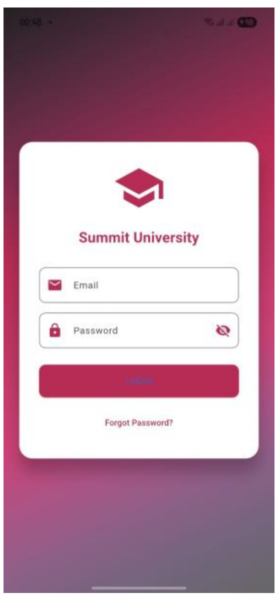

Figure 1.6: Finalized user interface

Designed and developed the user interface for the core application authentication layout, focusing specifically on the Student Login Screen. Built out the form containers using text fields for input values, designed submission button states, and managed custom interface spacing properties within a newly structured screens folder directory to handle presentation views separately from core logic files.

Tools/Software Used:

Android Studio, Flutter Framework, Dart Layout Widgets (Column, TextField, ElevatedButton).

Personal Reflection:

During the third week of developing my Flutter Student Management System, I focused on enhancing the application's features and improving its performance. I implemented additional functionalities such as student search, report generation, and data validation to ensure accurate record management. I also refined the user interface by improving navigation, layouts, and responsiveness across different screen sizes. Extensive testing was conducted to identify and fix bugs, ensuring that the system operated smoothly and efficiently.

Week 4:

Practical Activities Undertaken: